a Sentences and photographs pass through one pipeline

sentences: five relations and "a x and a y"

a tulip — part

a tulip on a barrel — composite

photographs: three layouts

part

composite

pretrained model
trained or random weights

part code $e(x)$

state h

target

readouts

tulip

barrel

part code $e(x)$

state h

target

readouts

fresh nouns and photograph classes never fit a readout

b Models across scale and training

scale	Pythia, 7 sizes from 160M to 12B
training	Pythia-1.4b, 6 checkpoints from step 0
other LLMs	Qwen3-8B, OLMo-2 7B and 13B, phi-4
text encoders	CLIP, T5, BERT, GPT-2 (11 models)
image encoders	CLIP, DINOv2, SigLIP (6 models)

every encoder also runs with random weights

c Do roles follow meaning or word order?

fit — a tulip on a barrel

test — a barrel under a tulip

figure readout returns

tulip: meaning

barrel: word order